# MODULE - 1.

* Rank of a Matrix = Order of largest non-Singular square sub matrix.

* Finding Rank using Echelon Form.
* Finding Rank using Gauss Elimination Method.
    - Augmented Matrix $[A:B]$ (combined Matrix)

* For a non-homogeneous linear System $(AX=B)$ :

CASE 1 : Rank $[A:B] \neq$ Rank $[A] \rightarrow$ Inconsistent $\rightarrow$ No Solution.

CASE 2 : Rank $[A:B]$ = Rank $[A]$ = No.of variables $\rightarrow$ consistent $\rightarrow$ Unique Solution.

CASE 3 : Rank $[A:B]$ = Rank $[A] <$ No.of variables $\rightarrow$ Consistent $\rightarrow$ Infinite Solutions.

* For a Homogeneous System of Equations $(AX=0)$

CASE 1 : $X=0$ [Null solutions / Trivial Solution]

CASE 2 : Rank $(A)$ = No.of variables $\rightarrow$ Trivial Solution (i.e $X=0$).

CASE 3 : Rank $(A) <$ No.of variables $\rightarrow$ Infinite Solutions (i.e Trivial & non-trivial)

* Characteristic Equation : $|A - \lambda I| = 0$

Eigen values : Roots of $|A - \lambda I| = 0$.

Spectrum : { Eigen values }

* SHORTCUTS :

(i) Characteristic Eq. (for 3x3 matrix) :

$$\boxed{\lambda^3 - (\text{Trace } A)\lambda^2 + (\text{Sum of cofactors of diagon}\ldots \text{ of } A)\lambda - |A| = 0}$$

(ii) characteristic Eq (2 x 2 Matrix) :

$$\boxed{\lambda^2 - (\text{Trace } A)\lambda + |A| = 0}$$

* Properties of Eigen Values :

(i) Sum of Eigen values = Trace A { Sum of Diagon

(ii) Product of Eigen values = $|A|$

(iii) $A, A^T \longrightarrow$ Same Eigen values

(iv) Eigen values (Diagonal Matrix) = Just the diagonal Elements.

(v) If $\lambda_1, \lambda_2 \ldots \lambda_n$ are eigen values of A, then

$\Rightarrow \frac{1}{\lambda_1}, \frac{1}{\lambda_2} \ldots\ldots \frac{1}{\lambda_n}$ are eigen values of $A^{-1}$

$\Rightarrow K\lambda_1, K\lambda_2 \ldots K\lambda_n$ " of $KA$

$\Rightarrow \lambda_1^n, \lambda_2^n \ldots\ldots \lambda_n^n$ " of $A^n$

$\Rightarrow \lambda_1^n + K, \lambda_2^n + K \ldots\ldots$ " of $A^n + KI$

* Eigen vector:

$$\boxed{(A-\lambda I)X = 0}$$

NOTES :

(i) Eigen Space of A : $E[\lambda] = \{$ Linearly independa[nt] Eigen vectors corresponding to $\lambda\}$

(ii) Eigen Basis = $\{X_1, X_2 \dots\}$ (set of all Eigen vectors)

(iii) Algebraic Multiplicity = No.of times eigen valu[e] $\lambda$ has repeated.

(iv) Geometric Multiplicity = No.of Linearly indepen[dent] Eigen vector corresponding to $\lambda$.

* $3^\circ$ polynomial → 3 roots

$2^\circ$ polynomial → 2 roots

* Diagonalization of Matrix :

$$\boxed{P^{-1}AP = D_{F}} \text{ -- Spectral Matrix}$$

Model Matrix, $P = \begin{bmatrix} X_1 & X_2 & \dots \end{bmatrix}$

- A Matrix is diagonalizable only if it :

(i) Have distinct Eigen values

(ii) No complex numbered eigen values.

(iii) For 2×2 to be diagonalizable, it must ha[ve] two linearly independent eigen vector

(iv) For 3x3 to be diagonalizable → Must hav[e]

three eigen vectors (linearly independent)

NOTE :

For $n \geq 0$, $n^{th}$ power of a square matrix A

is given by $\boxed{A^n = PD^nP^{-1}}$

* A real square matrix A is called :

(i) Symmetric if $A^T = A$

(ii) Skew Symmetric if $A^T = -A$

(iii) Orthogonal if $AA^T = I$ or $A^T = A^{-1}$

* Quadratic Form : (2nd degree order)

$$\boxed{Q = X^T A X}$$

→ Symmetric Matrix.

(i) Positive definite if all $\lambda$ is positive.

(ii) Positive semi-definite if at least one $\lambda$ is zer[o]

and rest are positive.

(iii) Negative definite if all $\lambda$ is negative.

(iv) Negative Semi-definite if at least one $\lambda$ is

zero and rest are negative.

(v) Indefinite if $\lambda$ includes both +ve & -ve.

(vi) Rank = No. of non-zero eigen values.

(vii) $\boxed{\text{Signature} = [n_0 \quad n_+ \quad n_-]}$

$n_-$ ---> No. of -ve $\lambda$

$n_0$: No. of zero $\lambda$ $n_+$: No. of +ve $\lambda$

## * Reduction of Quadractic Form to Canonical Form

- Symmetric Matrix A. [To to found]
- Find $\lambda_1, \lambda_2, \lambda_3$ & $X_1, X_2, X_3 \ldots$
- Normalized Model Matrix.

$$\boxed{P = \left[\frac{X_1}{\|X_1\|}, \quad \frac{X_2}{\|X_2\|} \quad \frac{X_3}{\|X_3\|}\right]}$$

Norm $\|X_i\| = \sqrt{\text{Sum of Squares}}$ column element

i.e $X_1 = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$ $\therefore$ $\|x_1\| = \sqrt{1^2 + 2^2 + (-3)^2} = \sqrt{14}$

- Verify $\boxed{P^{-1}AP = P^TAP = D}$
- Write Canonical form:

$$\boxed{[y_1 \quad y_2 \quad y_3] \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \lambda_1 y_1^2 + \lambda_2 \ldots \lambda_3 y_3^2}$$

## * Principal Axis Theorem [Using X = PY Substitution

# MODULE-2

* $f_x = \frac{\partial f}{\partial x}$ (with y as constant) ; $f_y = \frac{\partial f}{\partial y}$ (with $x$ as constant)

* $f_{xx} = \frac{\partial^2 f}{\partial x^2}$ ; $f_{yy} = \frac{\partial^2 f}{\partial y^2}$ ; $f_{xy} = \frac{\partial^2 f}{\partial y \partial x}$ ; $f_{yx} = \frac{\partial}{\partial}$

* $\frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}$

* Product rule : $\frac{duv}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$

* $f_{xy} = f_{yx}$ [ Same for all continuous functions

* Laplace Equation : $f_{xx} + f_{yy} = 0$

* Heat's Equation : $\frac{\partial z}{\partial t} = c^2 \frac{\partial^2 z}{\partial x^2}$

* Quotient Rule : $\frac{d(u/v)}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$

* Slope of Surface in [ $z = f(x,y)$ ]

(i) $x$ - $\text{dir}^n$ = $\frac{\partial z}{\partial x}$

(ii) $y$ - $\text{dir}^n$ = $\frac{\partial z}{\partial y}$

## Chained Rule :

**CASE 1 :** $z \rightarrow (x, y) \rightarrow t$

$$\boxed{\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}}$$

**CASE 2 :** $z \rightarrow (x, y) \rightarrow (u, v)$

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v}$$

* **Local Linear Approximation :**

$$\boxed{L(x, y) = f(a, b) + (x-a) f_x(a, b) + (y-b) f_y(a, b)}$$

* **Distance formula :** $[(x_1, y_1) \quad (x_2, y_2)]$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

* **Total Differentials :**

$$w = f(x, y, z)$$

$$\boxed{dw = f_x dx + f_y dy + f_z dz}$$

* $\frac{d}{dt} \sec x = \sec x \cdot \tan x$.

$\therefore \frac{d}{dt} \sec^2 x = \frac{d}{dt} (\sec x)^2 = 2 \sec x \, [\sec x \cdot \tan x]$

* $$\boxed{\text{Relative Error} = \frac{dz}{z} \qquad \text{Percentage Error} = \frac{dz}{z} \times 100}$$

* Area of Ellipse, $\boxed{A = \pi ab}$ → $b$ = Minor axis
  $a$ = Major axis

* Diagonal of rectangular box, $D = \sqrt{x^2 + y^2 + z^2}$

* **Implicit Differentiation:**

  (i) $f(x,y) = c$ ; defining $y$ implicity as a function of $x$ then,

  $$\boxed{\frac{dy}{dx} = -\frac{f_x}{f_y}}$$

  (ii) $f(x,y,z) = c$ ; defining $z$ implicity as a f of $x$ and $y$ then,

  $$\boxed{\frac{\partial z}{\partial x} = -\frac{f_x}{f_z} \; ; \; \frac{\partial z}{\partial y} = -\frac{f_y}{f_z}}$$

⇒ **Second Partial Test [For Relative Maxima & Minima]**

(i) Find $f_x$ & $f_y$

(ii) Solve $f_x = 0$ & $f_y = 0$ Simultaneously & find critical point

(iii) Find Discriminant, $D = f_{xx} \cdot f_{yy} - (f_{xy})^2$

(iv) Find $D$ & $f_{xx}$ at every critical point.

⇒ • If $D > 0$ & $f_{xx} > 0$ at $(a,b)$ then

$(a,b) \rightarrow$ Point of Relative Minima

• If $D > 0$ & $f_{xx} < 0$ at $(a,b)$ then,

$(a,b) \rightarrow$ Point of Relative Maxima.

• If $D < 0$ at $(a,b)$ then,

$(a,b) \rightarrow$ saddle point (No maxima nor Minima)

• If $D = 0$ at $(a,b)$ then,

NO CONCLUSIONS!!

* <u>Two point form</u> : A $(x_1, y_1)$ ——— B $(x_2, y_2)$

Equation of AB : $y - y_1 = \dfrac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

MODULE-3

* Integration by Parts :

$$\left(1^{st} \text{ fn} \times \int 2^{nd} \text{ fn}\right) - \int\left(\frac{d}{dx} 1^{st} \text{ fn} \times \int 2^{nd} \text{ fn}\right)$$

i.e $$\int f(x)\,g(x)\,dx = \left[f(x) \times \int g(x)\,dx\right] - \int\left[f'(x) \times \int g(x)\,d\right.$$

* $\sin n\pi = 0$

$\cos(n\pi) = (-1)^n$

$\cos(2n\pi) = (-1)^{2n} = 1$

* $$\int_a^b \int_c^d \int_e^f F(x,y,z)\,dx\,dy\,dz = \int_e^f f(x)\,dx \cdot \int_c^d g(y)\,dy \cdot \int_a^b h(z)$$

only: if a, b, c, d, e, f are constants &

if $F(x,y,z)$ can be written as product of

$f(x)$, $g(y)$, $h(z)$.

* Change of Order of Integration → Don't forget to change the limit.

* $e^{\infty} = \infty$

$e^{-\infty} = \frac{1}{e^{\infty}} = \frac{1}{\infty} = 0$

$\frac{1}{\infty} = 0$

$\frac{\infty}{a} = \infty$

$a + \infty = \infty$ (or $\infty + a = \infty$)

$a - \infty = -\infty$ (or $-\infty + a = -\infty$)

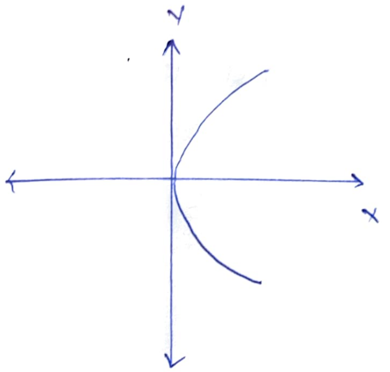
* $\int \frac{x}{x^2+y^2}\,dy = \tan^{-1}\left(\frac{y}{x}\right)$

$\int \frac{x}{x^2+y^2}\,dx = \frac{1}{2}\log|x^2+y^2|$

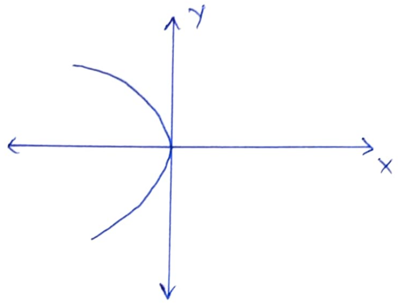
* $\log A - \log B = \log\frac{A}{B}$

$\log A + \log B = \log AB$

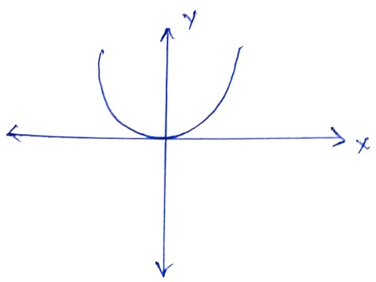
*



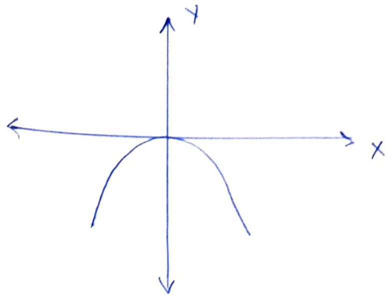

$\Rightarrow y^2 = 4ax$




$\Rightarrow y^2 = -4ax$




$\Rightarrow x^2 = 4ay$




$\Rightarrow x^2 = -4ay$

* Type 1 : $\int dy\,dx$

  Type 2 : $\int dx\,dy$

* Eq. of Circle : $(x-h)^2 + (y-k)^2 = r^2$

  $(h, k)$ = (centre coordinates)

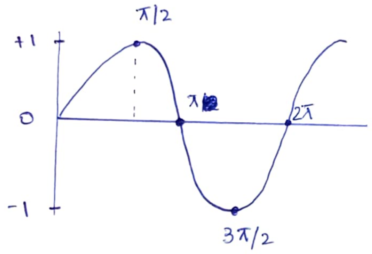
  $r$ = radius.

⇒ **Area Using Double Integral :**

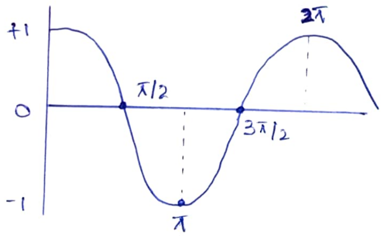
$$A = \iint_R dA = \iint_R dx\,dy = \iint_R dy\,dx$$

(Type 2) (Type 1)

* Sine :




Cosine :




* **Mass of Inhomogeneous Lamina :**

$$M = \iint_R \delta(x,y)\,dA$$

* **Centre of Gravity** : $(\bar{x}, \bar{y})$

$$\bar{x} = \frac{\iint_R x\,\delta(x,y)\,dA}{M} \; ; \quad \bar{y} = \frac{\iint_R y\,\delta(x,y)\,dA}{M}$$

* $y^2 = 4ax$ [Parabola]

$y = +\sqrt{4ax}$ [Implies positive region (1st Quadrant)]

$y = -\sqrt{4ax}$ [$\Rightarrow$ Negative region (IVth Quadrant)]

* **Volume Using Double Integral:**

$$\boxed{V = \iint_R z \, dA}$$

* $\int_{-a}^{a} \text{odd fn} = 0$

$$\int_{-a}^{a} \sqrt{a^2 - x^2}\, dx = \frac{\pi}{2} a^2$$

* **Intercept Form:** $\frac{x}{a} + \frac{y}{b} = 1$

$b$ ---> y-intercept

$a$ ---> x-Intercept

* **SHORTCUT:** Volume of Tetrahedron bounded by coordinate plane & plane $z = c - \frac{cx}{a} - \frac{cy}{b}$ is

$$\boxed{V = \frac{abc}{6}}$$

$\Rightarrow$ **Volume Using Triple Integral**

$$\boxed{V = \iiint_D dv = \iiint_D dx\,dy\,dz}$$

⇒ **Polar coordinates :**

$x = r\cos\theta$

$y = r\sin\theta$

$x^2 + y^2 = r^2$

$dA = r\,dr\,d\theta$

⇒ **Spherical coordinates :**

$x = r\sin\theta\cos\phi$

$y = r\sin\theta\sin\phi$

$z = r\cos\theta$

$dv = r^2\sin\theta\,dr\,d\theta\,d\phi$

$r^2 = x^2 + y^2 + z^2$

* $r$ : $0 \longrightarrow$ Given radius

* $\theta$ : $0 \rightarrow \pi$ ; for whole Sphere

  $0 \rightarrow \frac{\pi}{2}$ ; for Hemisphere

* $\phi$ : $0 \rightarrow 2\pi$ ; ALWAYS constant

# Unit ball means whole Sphere.

⇒ **Cylindrical coordinates :**

$x = r\cos\theta$

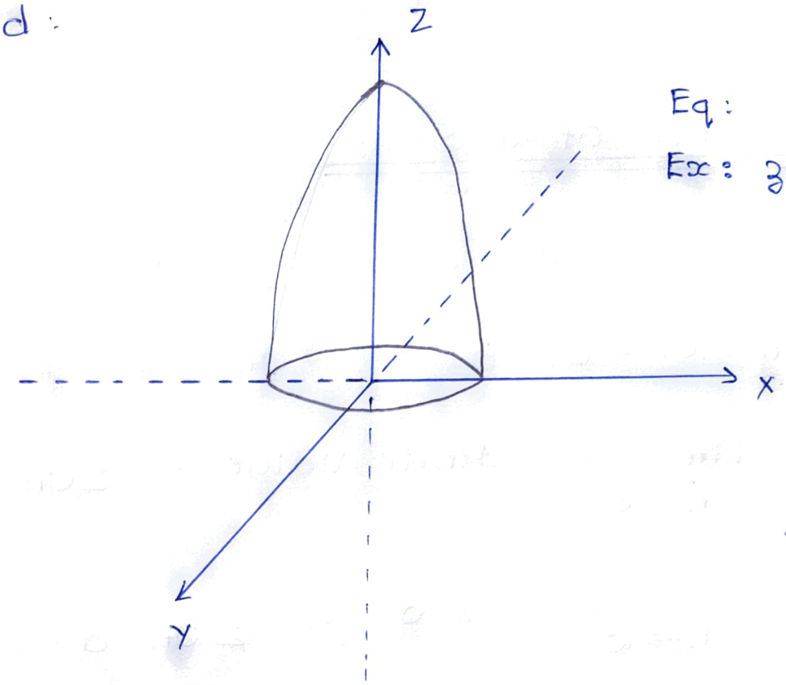
$y = r\sin\theta$

$z = z$

$x^2 + y^2 = r^2$

$dv = r\,dz\,dr\,d\theta$

* Paraboloid :




Eq :

Ex : $z = 9 - x$

# MODULE - 4

## ⇒ Convergence of a Series :

$$\Sigma U_n = U_1 + U_2 + U_3 \dots U_n$$

Let $S_n = U_1 + U_2 + U_3 + \dots U_n$

* $\lim\limits_{n\to\infty} S_n =$ Finite value $\Rightarrow \Sigma U_n$ converges

* $\lim\limits_{n\to\infty} S_n = \pm\infty \Rightarrow \Sigma U_n$ diverges.

## ⇒ Geometric Series :

$$\Rightarrow a + ar + ar^2 + \dots\dots, \quad a \neq 0$$

* Converges if $-1 < r < 1$
* Diverges otherwise
* If converges then sum, $S = \dfrac{a}{1-r}$

## ⇒ Limit Comparison test :

$\lim\limits_{n\to\infty} \dfrac{U_n}{V_n} \neq 0$, a finite values, then

$\Sigma U_n$ and $\Sigma V_n$ converge/diverge together

## ⇒ P- Series :

$$\sum_{n=1}^{\infty} \frac{1}{n^P} = \frac{1}{1^P} + \frac{1}{2^P} + \dots\dots$$

* Converges if $P > 1$
* Diverges if $0 < P \leq 1$

* Equation for finding $n^{th}$ term : $a + (n-1)d$

⇒ **Cauchy's Root test :**

$$\boxed{\lim_{K \to \infty} (U_K)^{1/K} = l}$$

* converges if $l < 1$
* Diverges if $l > 1$
* Test fails if $l = 1$

⇒ **D' Alembert's Ratio test :**

$$\boxed{\lim_{K \to \infty} \frac{U_{K+1}}{U_K} = l}$$

* converges if $l < 1$
* Diverges if $l > 1$
* Test fails if $l = 1$

→ **Leibnitz's Test :** An alternating series converg only if the following two condition satisfies :

(i) $U_1 > U_2 > U_3 > \ldots$

(ii) $\lim_{n \to \infty} U_n = 0$

→ **Absolute convergence and conditional convergen**

A series $\Sigma U_n$ is said to converge absolut if $\Sigma |U_n|$ is convergent.

If $\sum u_n$ converges but $\sum |u_n|$ diverges, th

series $\sum u_n$ is said to be conditionally

convergent.

⇒ **Ratio Test for Absolute convergence :**

$$\boxed{\lim_{k \to \infty} \left| \frac{u_{k+1}}{u_k} \right| = l}$$

* Converges absolutely if $l < 1$
* Diverges if $l > 1$
* Test fails if $l = 1$

⇒ **Comparison Test :**

* If $\sum u_n \le \sum V_n$ and if $\sum V_n$ converges, then $\sum u_n$ also converges.

* If $\sum u_n \le \sum V_n$ and if $\sum u_n$ diverges, then $\sum V_n$ also diverges.

# MODULE-5

⇒ **Taylor Series :**

$$f(x) = f(x_0) + \frac{(x-x_0)}{1!} f'(x_0) + \frac{(x-x_0)^2}{2!} f''(x_0) + \cdots$$

⇒ **Maclaurin Series :**

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \cdots\cdots$$

⇒ **Binomial Series :**

$$(1+x)^m = 1 + mx + \frac{m(m-1)x^2}{2!} + \frac{m(m-1)(m-2)x^3}{3!} + \cdots$$

⇒ **Fourier Series :**

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[ a_n \cos \frac{n\pi x}{l} + b_n \sin\left(\frac{n\pi x}{l}\right) \right]$$

# $a_0 = \frac{1}{l} \int_{\alpha}^{\alpha+2l} f(x)\,dx$ # $l = \frac{\text{upper} - \text{low}}{2}$

$$a_n = \frac{1}{l} \int_{\alpha}^{\alpha+2l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$b_n = \frac{1}{l} \int_{\alpha}^{\alpha+2l} f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

* $\sin(n\pi) = 0$

$\cos(n\pi) = (-1)^n$

$\cos(2n\pi) = (-1)^{2n} = 1$

$\cos(-\theta) = \cos\theta$

$\sin(-\theta) = -\sin\theta$

* $\int uv = uv_1 - u'v_2 + u''v_3 - \dots$

* $\int e^{ax}\cos bx\,dx = \dfrac{e^{ax}}{a^2+b^2}\left[a\cos bx + b\sin bx\right]$

$\int e^{ax}\sin bx\,dx = \dfrac{e^{ax}}{a^2+b^2}\left[a\sin bx - b\cos bx\right]$

⇒ * If $f(x)$ = Even fn in the interval $-l < x < l$,

then $b_n = 0$

* If $f(x)$ = Odd fn in the interval $-l < x < l$,

then $a_0 = 0$ & $a_n = 0$

* Test for even functions :

$f(-x) = f(x)$ ; even only if satisfied.

* $\int_{-l}^{l}$ Even fn $= 2\int_{0}^{l}$ Even fn

* $\int_{-l}^{l}$ odd fn $= 0$

* $f(x) = \begin{cases} x + 1, & -\pi < x < 0 \\ x - 1 & 0 < x < \pi \end{cases}$

→ Interval : $\{-\pi, \pi\}$

→ $x+1$, $x-1$ > Similar fn. (Somewhat)

→ Sign difference for constant term

→ ∴ It is odd. Hence $a_0 = 0$ & $a_n = 0$

(*) $f(x) = \begin{cases} 1-x , & -\pi < x < 0 \\ 1+x , & 0 < x < \pi \end{cases}$

→ Interval : $(-\pi, \pi)$

→ $\left.\begin{matrix} 1-x \\ 1+x \end{matrix}\right\rangle$ Similar

→ Sign difference for variable

→ ∴ Its even. Hence $b_n = 0$

⇒ (i) <u>Fourier Cosine Series</u>

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right)$$

where $a_0 = \frac{2}{l}\int_0^l f(x)\,dx$

$a_n = \frac{2}{l}\int_0^l f(x) \cos \frac{n\pi x}{l}\,dx$

$l$ = upper lim − lower limit

(ii) <u>Fourier Sine Series</u> :

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

where $b_n = \frac{2}{l}\int_0^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$

⇒ <u>Parseval's Identity</u> :

$$\frac{1}{l}\int_{-l}^{l} [f(x)]^2\,dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} [a_n^2 + b_n^2]$$